

Vedantu

Vedantu's Pride, India's Glory. Riddhesh Strikes Gold Again!

56th International Physics Olympiad (IPhO) 2026



Riddhesh Bendale 3 Year Program

All 5 Indian Students
Won Gold

India Finished at Joint
Rank #1 in the World!

India's Previous 5-Gold
Sweep at IPhO Came in 2018

IPhO 2026, Colombia

Vedantu Today. Olympian Tomorrow.

★ BEST PERFORMANCE IN ★ SCIENCE OLYMPIADS

A LEGACY OF EXCELLENCE. A FUTURE OF IMPACT.



CHAMPION RESULTS!



PROOF THAT
HARD WORK WINS!

2020 - 2025

5 YEARS. 11 OLYMPIADS. 16 MEDALS. ONE CHAMPION.

Vedantu | Master Every Concept

 **2 GOLD**  **IPhO**
International Physics Olympiad

 **2 GOLD**  **IBO**
International Biology Olympiad

 **1 GOLD**  **IOAA**
International Olympiad on Astronomy & Astrophysics

 **1 GOLD**  **IOAA (Jr)**
International Olympiad on Astronomy & Astrophysics Jr.

 **1 GOLD**  **APhO**
Asian Physics Olympiad

 **2 SILVER**  **IOAA**
International Olympiad on Astronomy & Astrophysics

 **1 SILVER**  **APhO**
Asian Physics Olympiad

 **1 BRONZE**  **IOAA (Jr)**
International Olympiad on Astronomy & Astrophysics Jr.

 **1 GOLD**  **IJSO**
International Junior Science Olympiad

 **1 SILVER**  **IJSO**
International Junior Science Olympiad

 **2 SILVER**  **ICho**
International Chemistry Olympiad



Nidhi Prashar

Chemistry



- 16+ Years of teaching experience
- Gold Medallist in the Master's Degree (Chemistry), UGC NET Qualified
- IJSO EXPERT with 150+ NSEJS Selections
- Mentored Science Olympiads GOLD MEDALISTS
- Teaching and mentoring students IIT, NEET, NSEC, IJSO exams
- Produced Single and double digit ranks in JEE/NEET (AIR 4 in AIIMS)

Examination Schedule

NSEA (Astronomy)	Saturday 21.11.26	2.30 pm to 4.30 pm
NSEP (Physics)	Sunday 22.11.26	8.30 am to 10.30 am
NSEC (Chemistry)	Sunday 22.11.26	11.30 am to 1.30 pm
NSEB (Biology)	Sunday 22.11.26	2.30 pm to 4.30 pm
NSEJS (Junior Science)	Sunday 22.11.26	2.30 pm to 4.30 pm

Telegram Channel



Mole Concept

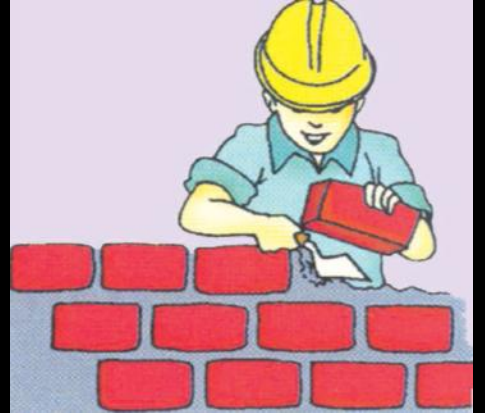


Highest weightage Chapter



Atom

An atom is the smallest particle of an element that can take part in a chemical reaction.



Just as bricks are the building blocks of all the houses, atoms are the building blocks of all matter.

Atomic mass Unit (amu)

The quantity $\left[\frac{1}{12} \times \text{mass of an atom of C-12} \right]$ is known as **atomic mass unit**

$$1 \text{ amu} = \frac{1}{12} \text{ mass of one atom } {}^{12}\text{C}$$

RAM

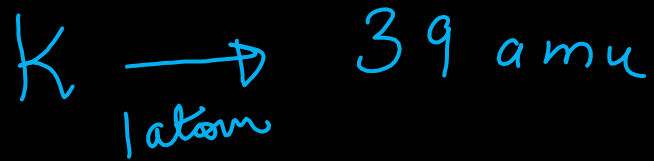
Unified mass (u)

Dalton

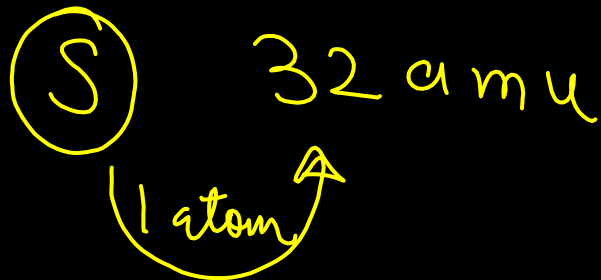
Atomic mass Unit

$$\text{Relative atomic mass} = \frac{\text{Mass of an atom of the element}}{\frac{1}{12} \times \text{Mass of an atom of carbon (C-12)}}$$

Element name	Atomic number	Atomic mass	Element name	Atomic number	Atomic mass
Hydrogen	1	1	Sodium	11	23
Helium	2	4	Magnesium	12	24
Lithium	3	7	Aluminium	13	27
Beryllium	4	9	Silicon	14	28
Boron	5	11	Phosphorus	15	31
Carbon	6	12	Sulphur	16	32
Nitrogen	7	14	Chlorine	17	35.5
Oxygen	8	16	Argon	18	40
Fluorine	9	19	Potassium	19	39
Neon	10	20	Calcium	20	40



1 atom ka mass, agar amu mein
blaya jaye toh usse atomic
mass of element kehte hain!



Mole

A mole is the amount of substance that contains as many species (atoms, molecules, ions or other particles) as there are atoms in exactly 12 gm of C-12

$$\text{Avogadro's No} = N_A$$

$$1 \text{ mole} = \underline{6.022 \times 10^{23}} \text{ species}$$

Mole

👉 1 mole of any particle = 6.022×10^{23} in numbers

Same as,

👉 1 dozen = 12 numbers, 1 gross = 144 numbers



1 dozen of egg



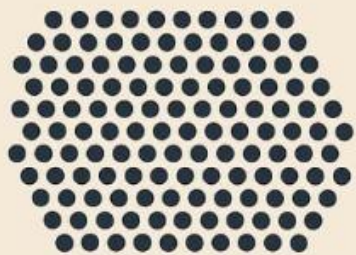
1 gross of smiley

The exact value of Avogadro's number is:

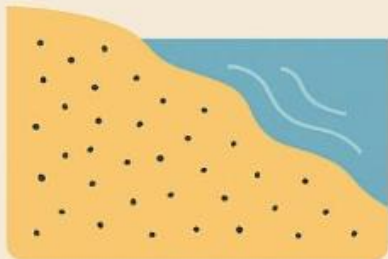
$$6.022\,140\,76 \times 10^{23} \text{ mol}^{-1}$$

This value was defined in 2019 by the International System of Units (SI) as an exact constant.

Mole Concept Visualization



= 1 mole = $6,022 \times 10^{23}$ particles



grains of sand
on Earth



stars in
the galaxy



drops of water
in the ocean

"What if you distribute 1 mole of rupees in total to every person on Earth, how much each person would get?"

**With ~8 billion people, each person would get
~75 quadrillion rupees!**

↓
 10^{15}

If you counted 1 number per second, how long to reach Avogadro's number?"

Over 19 million years!

Bank employee

5 kg

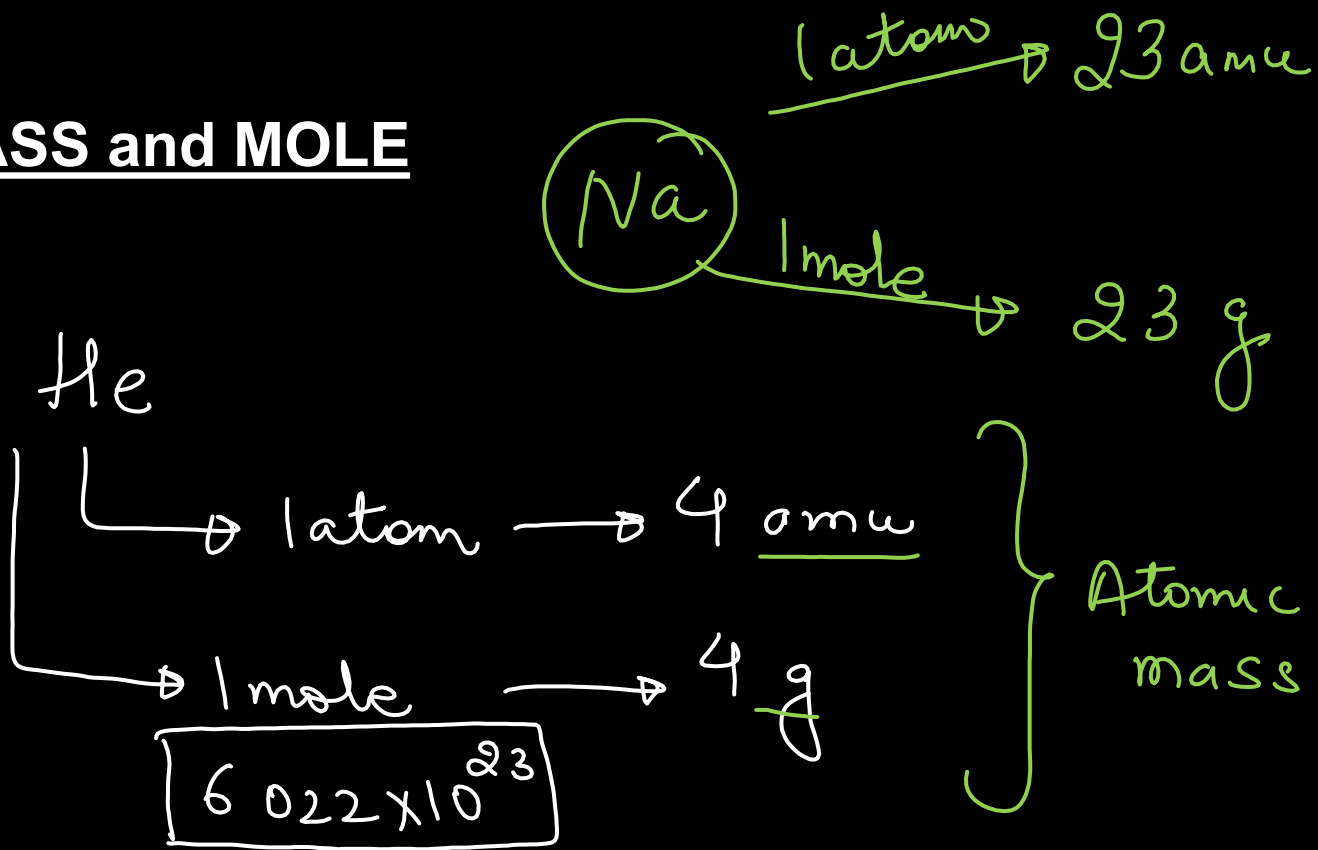
$$\frac{5000g}{200}$$

$$\begin{array}{r} \text{No of coins} \\ 10 \\ \hline \end{array} \downarrow \begin{array}{r} 200 \text{ gms} \\ \hline \end{array}$$

$$\begin{array}{l} 200g \rightarrow 10 \text{ coins} \\ 5000g \rightarrow \frac{10}{200} \times 5000 \\ \quad \quad \quad = 250 \end{array}$$

1 Rupee
coins

MASS and MOLE



RELATION BETWEEN AMU AND GRAMS

mass of 1 H-atom = 1 amu

" " 10 " = 10 amu

" " 100 " = 100 amu

" " N_A = N_A amu = 1 g

$$N_A = 6.022 \times 10^{23}$$

$$\Rightarrow 1 \text{ amu} = \frac{1}{N_A} \text{ g}$$

Note

$$\Rightarrow 1 \text{ amu} = 1.66 \times 10^{-24} \text{ g}$$

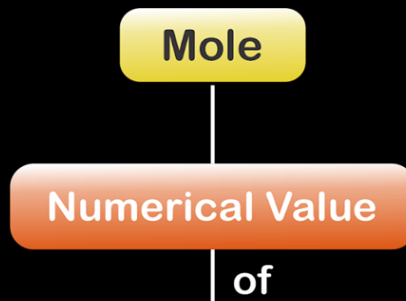
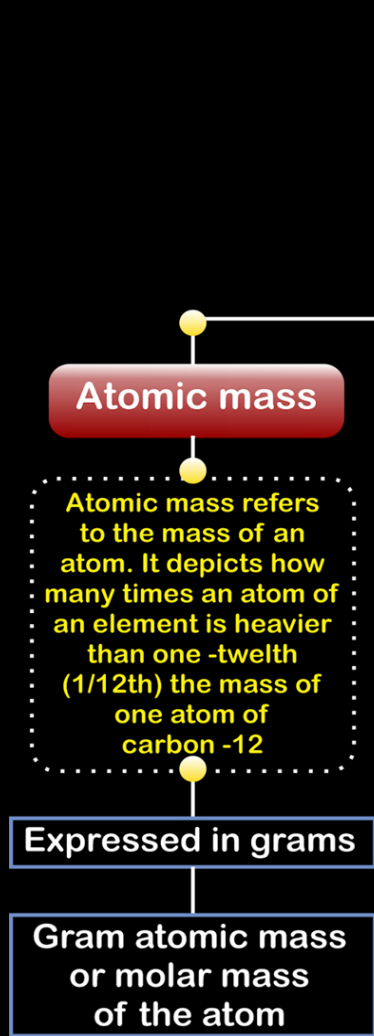
Vedantu

For example:

$$\begin{aligned}\text{Mass of an atom of hydrogen} \\ &= 1.6736 \times 10^{-24} \text{ g}\end{aligned}$$

Thus, in terms of amu, the mass

$$\begin{aligned}\text{of hydrogen atom} &= \frac{1.6736 \times 10^{-24} \text{ g}}{1.66056 \times 10^{-24} \text{ g}} \\ &= 1.0078 \text{ amu} \\ &= 1.0080 \text{ amu}\end{aligned}$$



56g He

1 mole = 4 g

56g = 14 moles of atom

$= 14 \times 6.022 \times 10^{23}$ atoms

56 amu He

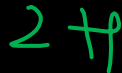
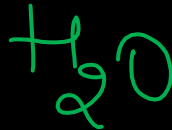
1 atom = 4 amu

No. of atom = $\frac{56}{4} = 14$ atoms

MOLECULES

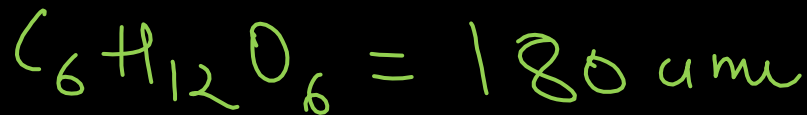
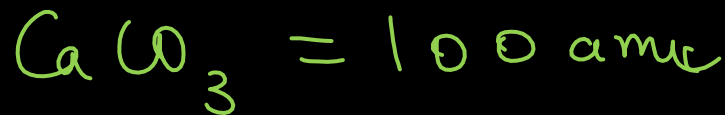
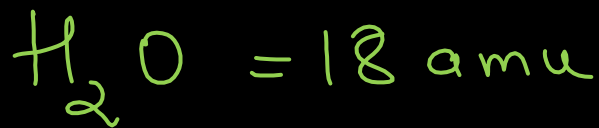


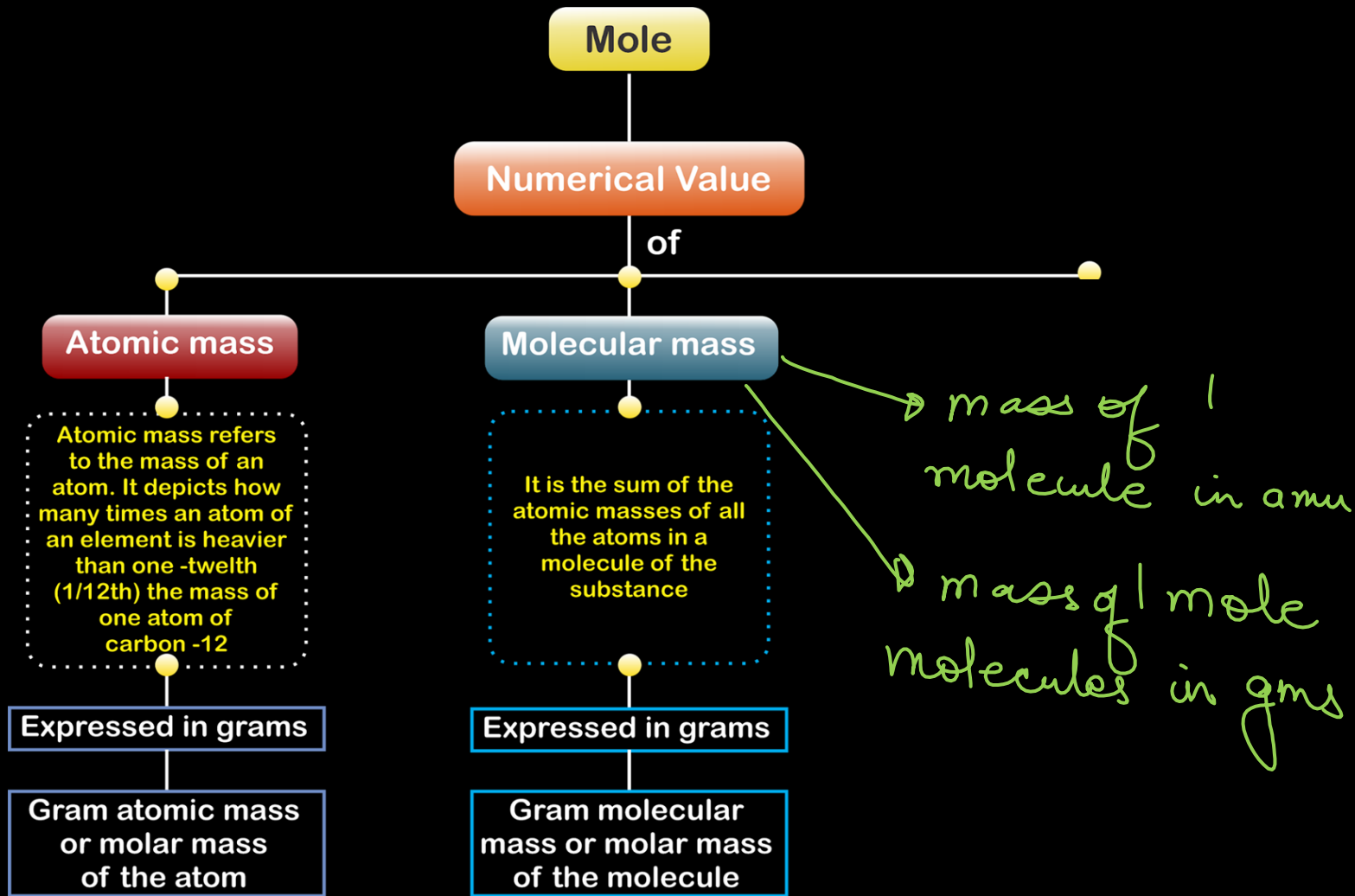
group of atoms

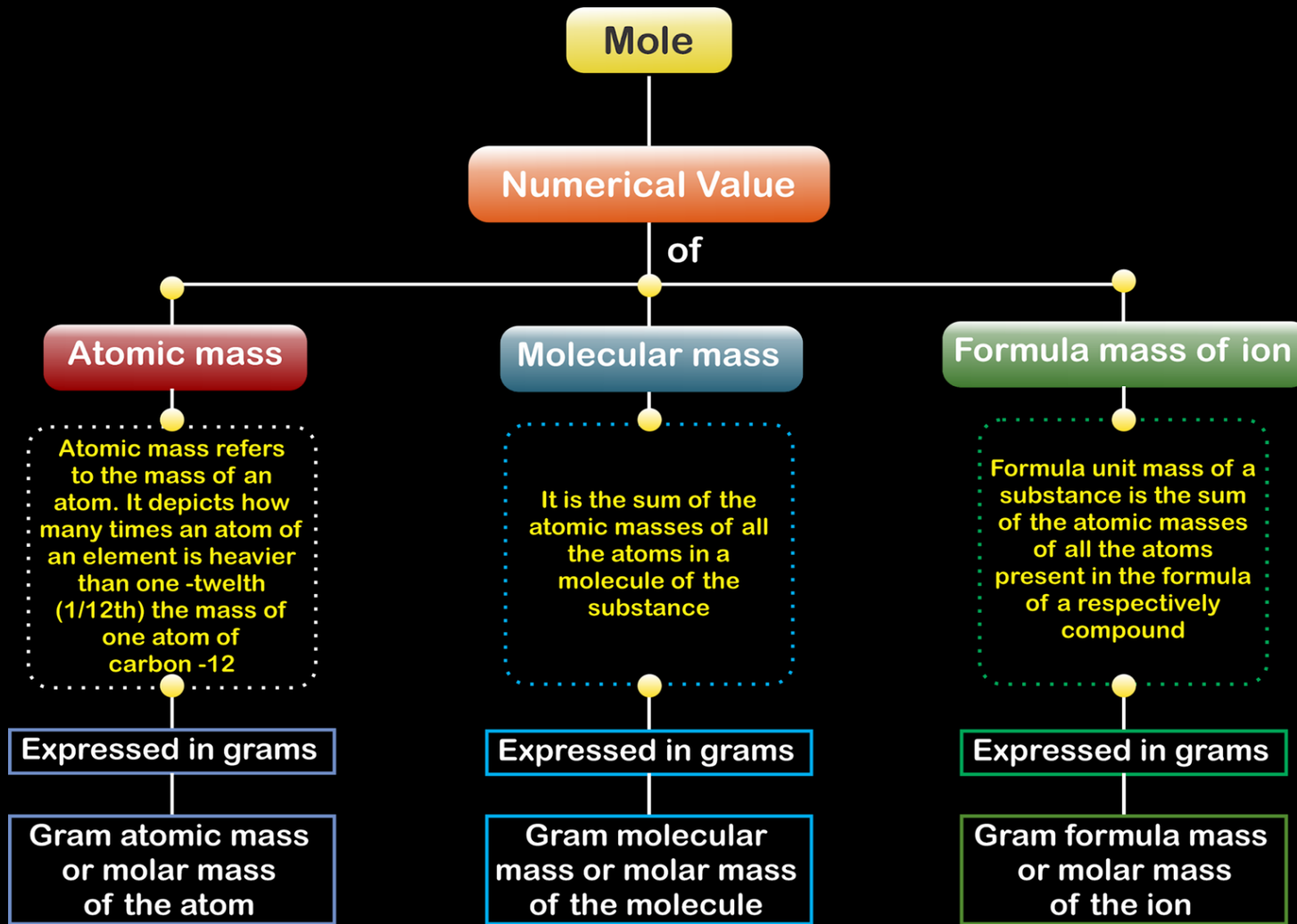




$$12 \text{ amu} + 2 \times 16 \text{ amu} = 44 \text{ amu}$$







For Atomic substances

❑ 1 Mole = Gram atomic mass = 6.022×10^{23} atoms of the element.

For molecular substances

□ 1 Mole = Gram molecular mass = 6.022×10^{23} molecules of the substance.

Method to calculate mole

From number of particles:

Most Important

$$\text{Number of moles} = \frac{\text{Given number of particles}}{\text{Avogadro number}}$$

Method to calculate mole

From given mass:

most important

$$\text{Number of moles} = \frac{\text{Given mass}}{\text{Molar mass}}$$

460 g Na

$$\text{No of moles} = \frac{460}{23} = 20 \text{ moles of Na atoms}$$

36 g H_2O

$$\text{No of moles} = \frac{36}{18} = 2 \text{ mole}$$

$$= 2 \times 6.022 \times 10^{23}$$

→ mass of 1 mole H_2O molecules

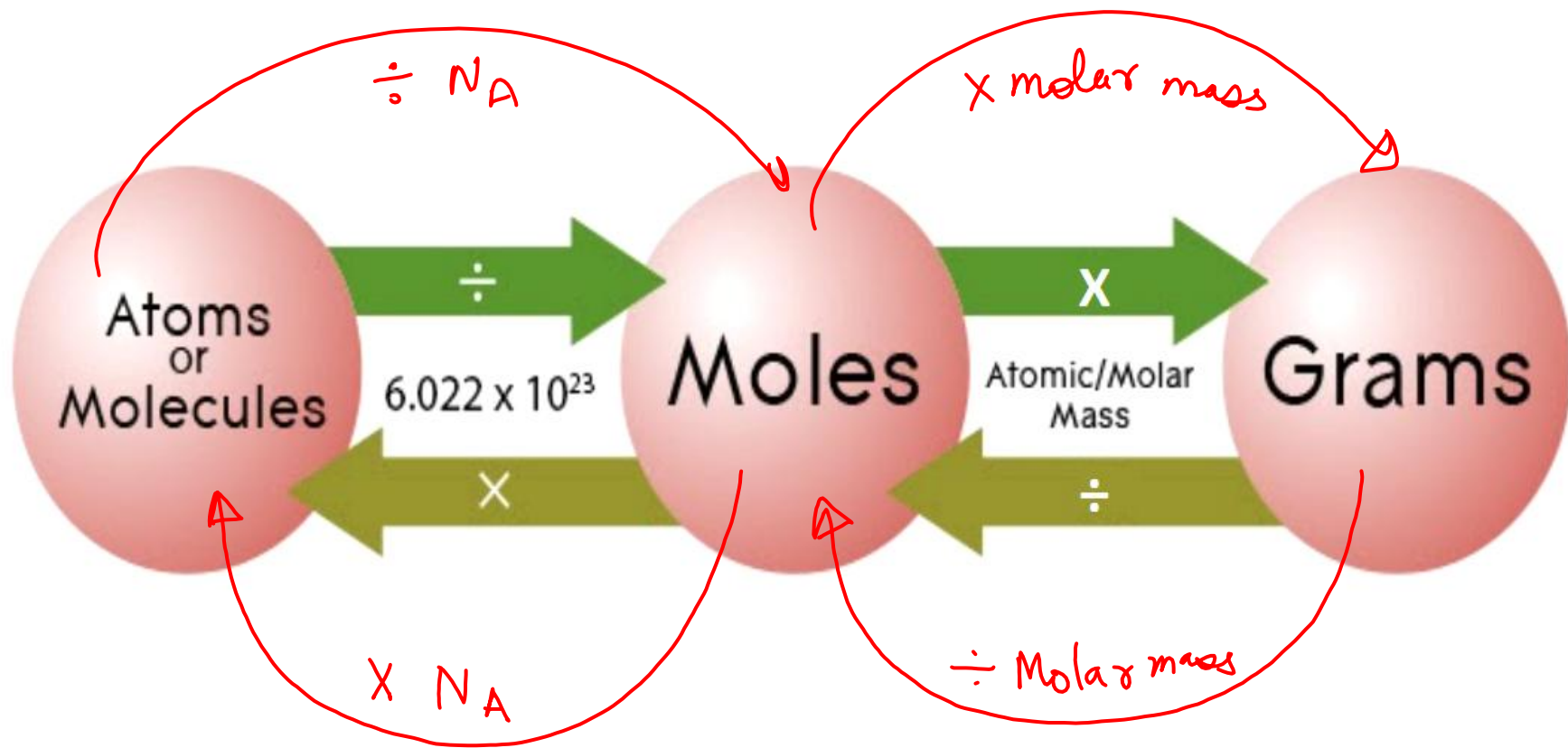
Method to calculate mole

From the given volume of gas:

$$\text{Number of moles} = \frac{\text{Volume given at STP}}{22.4 \text{ L}}$$

To find total number of identities

$$\text{The number of atoms} = \frac{\text{Given mass of atom}}{\text{Molar mass of atom}} \times \text{Avogadro number}$$



Q:1- What is the number of moles present in 52 g of He?

$$n = \frac{52}{4}$$

$$= \underline{13 \text{ moles}}$$

Solution:

Atomic mass of He = 4 u
Molar mass of He = 4g/mole

Thus, the number of moles

$$\frac{\text{Given mass}}{\text{molar mass}}$$
$$= \frac{m}{M} = \frac{52}{4} = 13$$

Q:2-The mass of one molecules of phosphorus (^{31}P) is

NSEJS 2010-11

A

31g

B

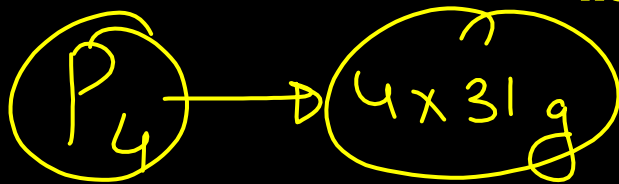
$2.050 \times 10^{-22}\text{g}$

C

$5.147 \times 10^{-23}\text{g}$

D

124g



1 mole molecule =

$$\frac{6022 \times 10^{23}}{1} = \frac{4 \times 31\text{g}}{6022 \times 10^{23}}$$

Q:2-The mass of one molecules of phosphorus (^{31}P) is

The mass of one molecules of phosphorus (^{31}P) is

A

31g

B

$2.050 \times 10^{-22}\text{g}$

C

$5.147 \times 10^{-23}\text{g}$

D

124g

Q:2-The mass of one molecules of phosphorus (^{31}P) is

NSEJS 2010-11

A

31g

B

$2.050 \times 10^{-22}\text{g}$

C

$5.147 \times 10^{-23}\text{g}$

D

124g

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31g

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$2.050 \times 10^{-22}\text{g}$

C

$5.147 \times 10^{-23}\text{g}$

D

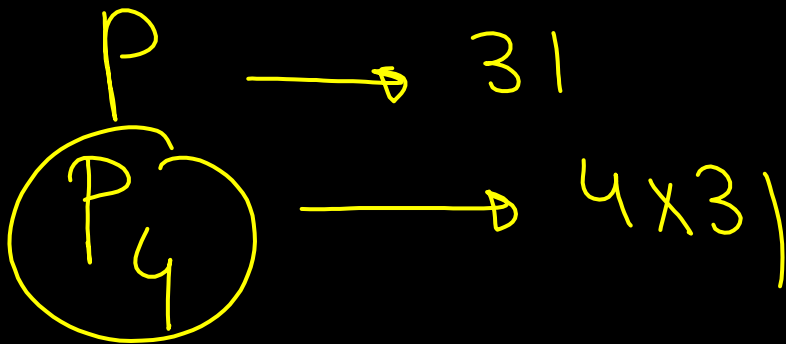
124g

Solution :

Mass of 6.023×10^{23} moles of Phosphorus = $31 \times 4\text{g}$

Mass of 1 molecules of phosphorus

$$= \frac{31 \times 4}{6.023 \times 10^{23}} = 2.050 \times 10^{-22}\text{g}$$



Q:3- Calculate the mass of 3.011×10^{23} number of N-atoms.

N

$$1 \text{ mole} = 6.022 \times 10^{23} = 14 \text{ g}$$

$$3.011 \times 10^{23} = \frac{14}{6.022 \times 10^{23}} \times 3.011 \times 10^{23}$$

Solution:

The number of moles (n)

$$\frac{\text{Given number of particles}}{\text{Avogadro number}} = \frac{N}{N_0}$$
$$= \frac{3.011 \times 10^{23}}{6.022 \times 10^{23}}$$

$$\Rightarrow m = M \times n$$
$$= 14 \times \frac{3.011 \times 10^{23}}{6.022 \times 10^{23}}$$
$$= 14 \times 0.5$$
$$= 7\text{g} \quad \checkmark$$

Q:4-The mass of 0.2 mole of Oxygen molecule is



NSEJS 2013-14

A 6.4g

B 3.2g

C 1.6g

D 2.75g

O_2

$$1 \text{ mole} = 32 \text{ g}$$
$$0.2 \text{ moles} = 32 \times 0.2 \text{ g}$$

Q:4-The mass of 0.2 mole of Oxygen molecule is

NSEJS 2013-14

The mass of 0.2 mole of Oxygen molecule is

A

6.4g

B

3.2g

C

1.6g

D

2.75g

$$\text{O}_2$$
$$\underline{1 \text{ mole}} = 32 \text{ g}$$

Solution :

$$\text{Mole} = \frac{\text{Given mass}}{\text{Gram molecular mass}}$$

$$0.2 = \frac{W}{32}$$

$$W = 6.4 \text{ g}$$

Q:5-Number of molecules present in a given amount of CO_2 and O_2 is the same at STP. If the weight of CO_2 is 5.5g, then the weight of O_2 is .

NSEJS 2010-11

A 5.5g

B 4.4g

C 4.0g

D 2.0g

At STP,

22.7 lt

1 mole gas = 22.4 lt

Q:5-Number of molecules present in a given amount of CO_2 and O_2 is the same at STP. If the weight of CO_2 is 5.5g, then the weight of O_2 is

NSEJS 2010-11

Number of molecules present in a given amount of CO_2 and O_2 is the same at STP. If the weight of CO_2 is 5.5g, then the weight of O_2 is

A

5.5g

B

4.4g

C

4.0g

D

2.0g

Solution :

If the no. of moles are same then no. of molecules also same

$$n_{\text{CO}_2} = \frac{5.5}{44}$$

$$= \underline{0.125 \text{ mole}}$$

$$1 \text{ mole of O}_2 = 32\text{g}$$

$$0.125 \text{ mole of O}_2 = 32 \times \underline{0.125} = \underline{4\text{g}}$$

Q: 6-8 grams of oxygen at NTP contain

O_2 $\downarrow$
same as STP

NSEJS 2012-13

A

1.5×10^{23} molecules

B

3.0×10^{23} molecules

C

6.023×10^{23} molecules

D

1.5×10^{22} molecules

$O_2 \rightarrow \boxed{32g}$
 $\downarrow$
 6.022×10^{23}
molecule

Q: 6–8 grams of oxygen at NTP contain

NSEJS 2012-13

8 grams of oxygen at NTP contain

A

1.5×10^{23} molecules

B

3.0×10^{23} molecules

C

6.023×10^{23} molecules

D

1.5×10^{22} molecules

Solution :

$$n = \frac{8}{32}$$

$$= \frac{1}{4}$$

$$\therefore 1 \text{ mole} = 6.023 \times 10^{23} \text{ molecule}$$

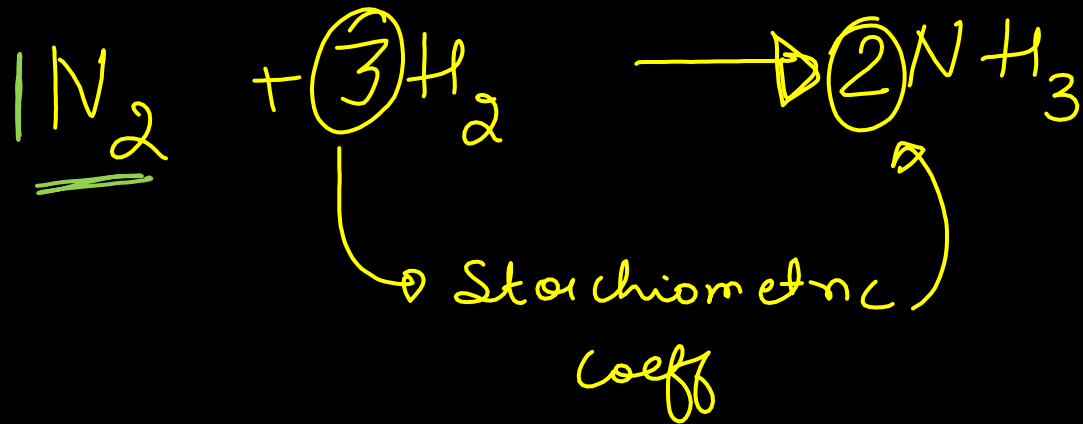
$$\therefore 1/4 \text{ mole} = 6.023 \times 10^{23} \times \frac{1}{4}$$

$$= 1.5 \times 10^{23} \text{ molecules}$$

Stoichiometry



Evergreen/entry Evergreen/entry



1 molecule	3 molecule	2 molecule
10 "	30 "	20 "
100	300	200
1 mole	3 mole	2 mole
$\textcircled{28 \text{ g}}$	$2 \times 3 = \textcircled{6 \text{ g}}$	$2 \times 17 \text{ g} = 34 \text{ g}$

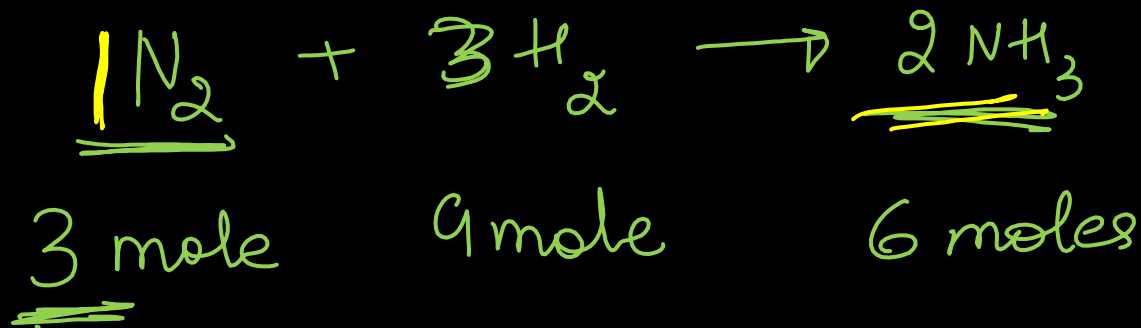
Particular	$2 \text{SO}_2 + 1 \text{O}_2 \rightleftharpoons 2 \text{SO}_3$		
Molecules	2 molecules	1 molecules	2 molecules
Atoms	2 S atom 4 O atom	2 O atom	2 S atoms 6 O atoms
Mass (AMU)	$2 \times 64 \text{ u} = 128 \text{ u}$	$1 \times 32 \text{ u} = 32 \text{ u}$	$2 \times 80 \text{ u} = 160 \text{ u}$
Mol Molecules	<u>2 mol molecules</u>	<u>1 mol molecules</u>	<u>2 mol molecules</u>
Mol Atoms	2 mol S atoms 4 mol O atoms	2 mol O atoms	2 mol S atoms 6 mol O atoms
Mass (Grams)	$2 \times 64\text{g} = 128\text{g}$	$1 \times 32\text{g} = 32\text{g}$	$2 \times 80\text{g} = 160\text{g}$

160g

Stoichiometry

The general method of calculation for stoichiometric problems:

- ☐ *Write down the balanced chemical equation.*
- ☐ *Write the relative number of moles or the relative masses(gram atomic or molecular masses) of the reactants and the products below the formulae.*
- ☐ *Now apply the unitary method to make the required calculations.*



4 mole 12 mole 8 moles

(LR) → 2 moles

8 - 6 = 2 mole left
8 moles

→ (ER)

(8 × 17) g
?

4 moles

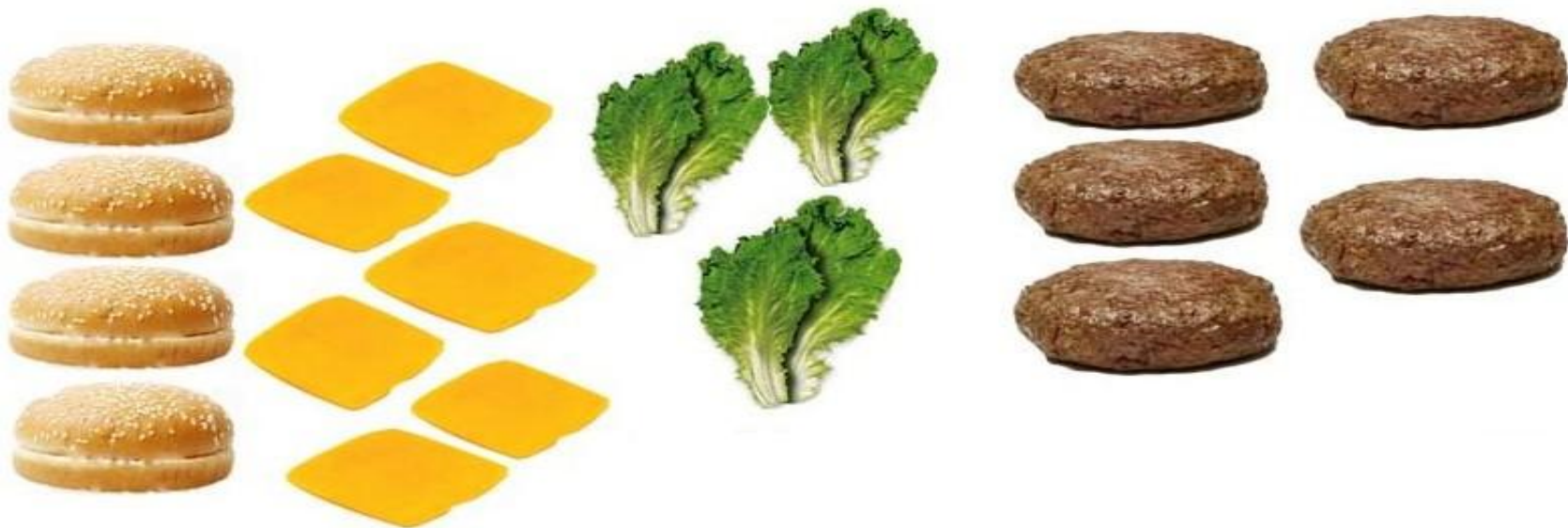
decides
(LR)

Limiting and Excess reagent

The Parts of a Burger



Now how many burger can be made



Three Only



- ❑ *Despite there being 4 buns, 6 slices of cheese, and 5 patties, we can only make 3 burgers because there are only 3 lettuce leafs.*
- ❑ *Hence lettuce leaf here known as **limiting reagent** and all other ingredients left over are known as **excess reagent**.*

Limiting and Excess reagent

Limiting Reagent (LR)

The reactant that gets **exhausted** first and limits or stops the reaction is called as **limiting reagent**

Excess Reagent (ER)

The **reactant that is left** at the end of the reaction is called as **excess reagent**

SIRF Method and k Ratio

$$k = \frac{\text{Given Moles}}{\text{Stoichiometric}}$$

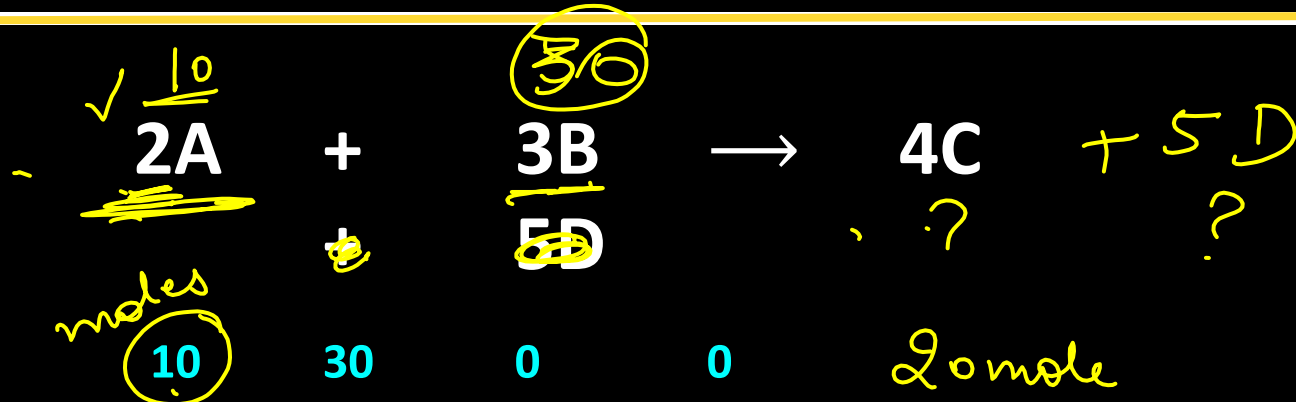
S = Stoichiometric
Coefficient I = Initial
Amount
R = Reacted

Reactant for which k is minimum is the LR

$$\text{Moles Reacted} \quad k_{\min} \times \text{Stoich Coeff}$$

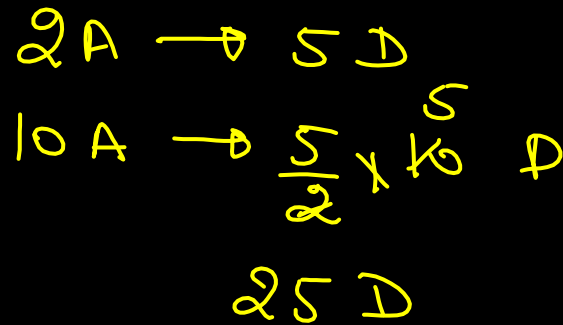
How to find Limiting Reagent

Initial moles



Determining the value of k for A and B

	A	B
k	$\frac{10}{2}$	$\frac{30}{3}$
	5 ✓	10
	↓	
	LR	



What mass of CO_2 will be formed when 6g of carbon is burnt in 32g of oxygen?

NTSE -2015

A

38g

B

12

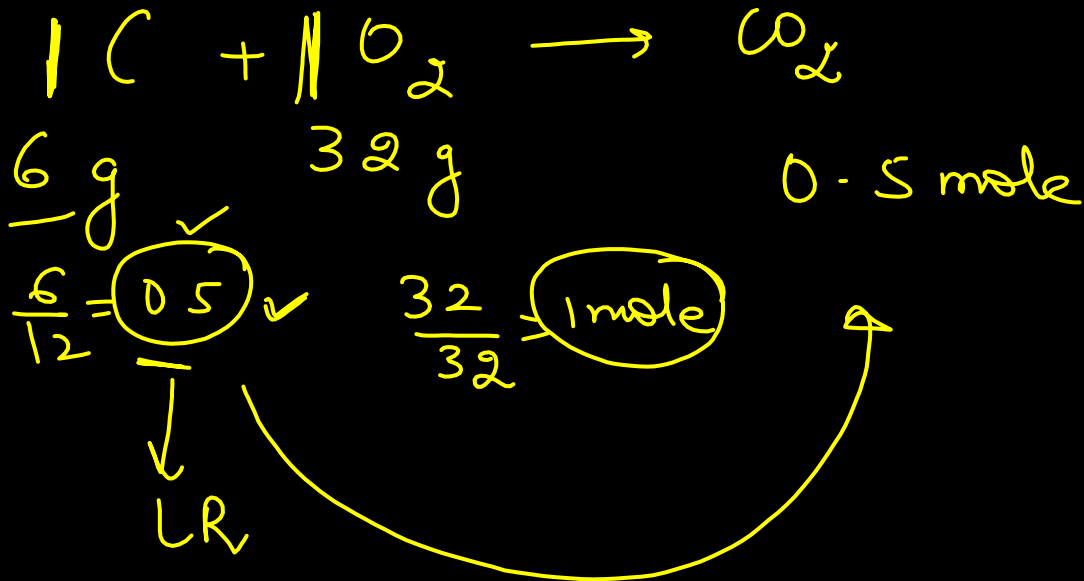
g mole

C

2

D

22g



What mass of CO_2 will be formed when 6g of carbon is burnt in 32g of oxygen?

A 38g

B 12
g

C 2

D 22g

Solution:

Given,

amount of C = 6g

amount of oxygen = 32g



$\therefore$ 12g of C gives the amount of $\text{CO}_2 = 44\text{g}$

$\therefore$ 1g of C gives the amount gm of $\text{CO}_2 = \frac{44}{12} \text{ g}$

$\therefore$ 6g of C gives the amount gm of CO_2

$$= \frac{44}{12} \times 6 = 22\text{g CO}_2$$

How much MgO is obtained on heating 5gm of (MgCO₃) Magnesium Carbonate-

Molar mass
 $24 + 12 + 3 \times 16 =$

NTSE -2013 (U.P)

A

2.4gm

B

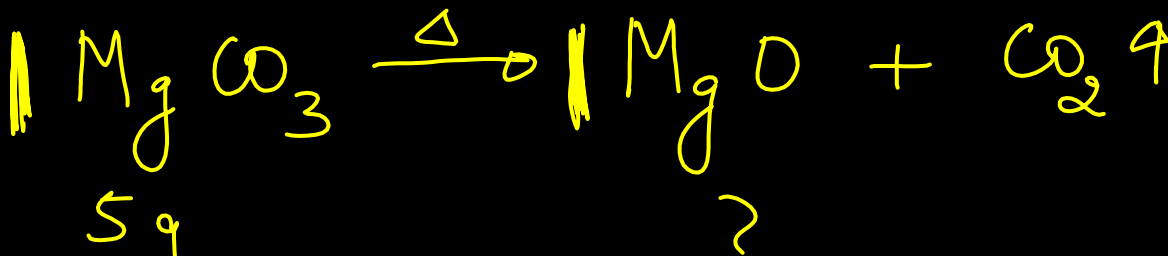
2.38gm

C

2.8g

D

3.28gm



5g

mole = $\frac{5}{84}$

How much MgO is obtained on heating 5gm of (MgCO_3) Magnesium Carbonate-

A

2.4gm

B

2.38gm

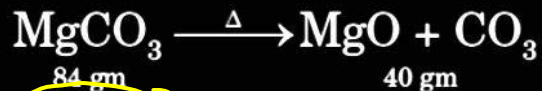
C

2.8g

D

3.28gm

Solution :



84 gm of MgCO_3 on decomposition gives 40 gm MgO

5 gm of MgCO_3 on decomposition gives

$$= \frac{40}{84} \times 5$$
$$= 2.38 \text{ gm}$$

✓ 2.38 gm $\rightarrow$ actual

5g
↓
2.38g MgO

Percentage Yield

In general, when a reaction is carried out in the laboratory we do not obtain the theoretical amount of product. The amount of product that is actually obtained is called the actual yield. Knowing the actual yield and theoretical yield, the % yield can be calculated by the following formula-

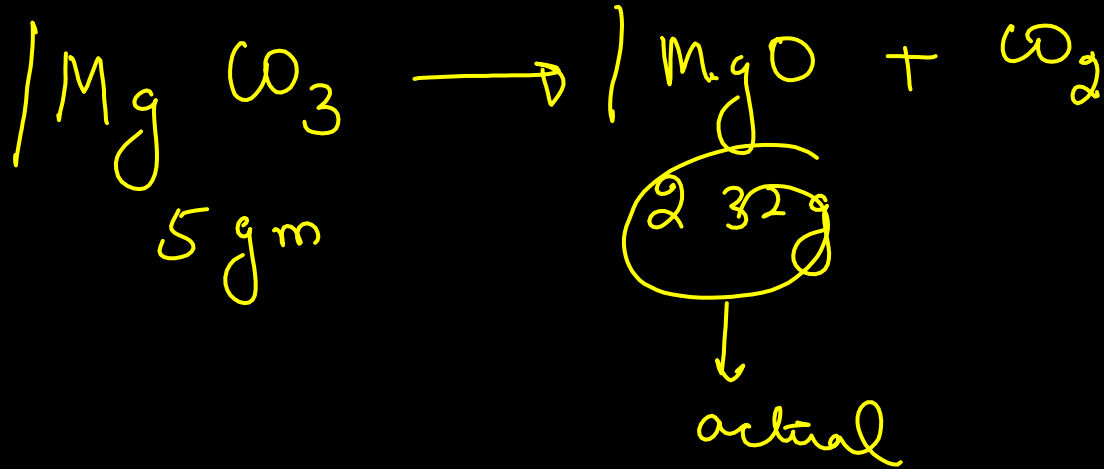
$$\text{Percentage yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100\%$$

↓
Calculations

Percentage Yield

SUBSCRIBE

Percentage Yield



$$\% \text{ yield} = \frac{232}{238} \times 100$$

What is the number of iron atoms present in a piece of iron weighing 2.8g? (Atomic mass of Fe = 56 u)

A

$$3.011 \times 10^{24}$$

B

$$5.011 \times 10^{23}$$

C

$$6.011 \times 10^{22}$$

D

$$3.011 \times 10^{22}$$

What is the number of iron atoms present in a piece of iron weighing 2.8g? (Atomic mass of Fe = 56 u)

A

$$3.011 \times 10^{24}$$

B

$$5.011 \times 10^{23}$$

C

$$6.011 \times 10^{22}$$

D

$$3.011 \times 10^{22}$$

Solution:

1 mole of Fe = 56 g

Now, 56 g of Fe contains = 6.022×10^{23} atoms

$$\begin{aligned}\text{So, 2.8 g of Fe contains} &= \frac{6.022 \times 10^{23}}{56} \times 2.8 \\ &= 3.011 \times 10^{22} \text{ atoms}\end{aligned}$$

When a certain amount of octane, C_8H_{18} , is burnt completely, 7.04 g CO_2 is formed. What is the mass of H_2O formed simultaneously?

A

1.62 g

B

6.48 g

C

3.24 g

D

2.28 g

When a certain amount of octane, C_8H_{18} , is burnt completely, 7.04 g CO_2 is formed. What is the mass of H_2O formed simultaneously?

A

1.62 g

B

6.48 g

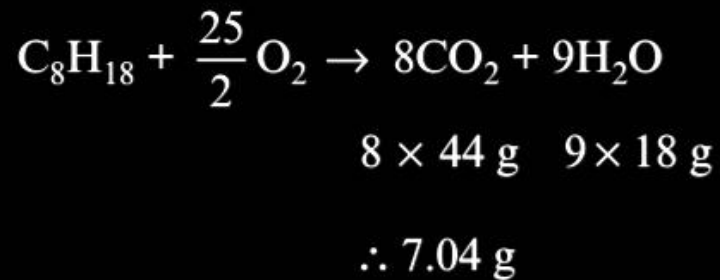
C

3.24 g

D

2.28 g

Solution:



$$\frac{9 \times 18}{8 \times 44} \times 7.04 = 3.24 \text{ g}$$

The minimum mass of sulphuric acid needed for dissolving 3 g of magnesium carbonate is

A 3.5 g

B 7.0 g

C 1.7 g

D 17.0 g

The minimum mass of sulphuric acid needed for dissolving 3 g of magnesium carbonate is

A 3.5 g

B 7.0 g

C 1.7 g

D 17.0 g

Solution:



84 g 98 g

$$\therefore 3.0 \text{ g} \quad \frac{98}{84} \times 3 = 3.5 \text{ g}$$

PERCENTAGE PURITY



What is Percentage Purity ?

Typical source of reactants are ORES and MINERALS. These are impure form of reactants

The actual amount participating in the reaction is less than the amount taken

Amount of product formed is calculated from actual amount of reactant

$$\% \text{ age Purity} = \frac{\text{Pure Mass of Reactant}}{\text{Impure Mass of Reactant}} \times 100$$

Gold Purity in Jewelry

Suppose you buy a gold ring that weighs 50 grams, and it's labeled as 18-karat gold.

Gold purity is usually measured in karats

24-karat gold is pure gold (100% pure)

18-karat gold means it's 18 parts gold out of 24

Let's Solve

Ex-1. Calculate the amount of lime (CaO) that can be prepared by heating 200 kg of limestone (CaCO₃) that is 95% pure.

190 kg



1 mole

1 mole

100 g $\longrightarrow$ 56 g

$200 \times 10^3 \text{ g CaCO}_3$

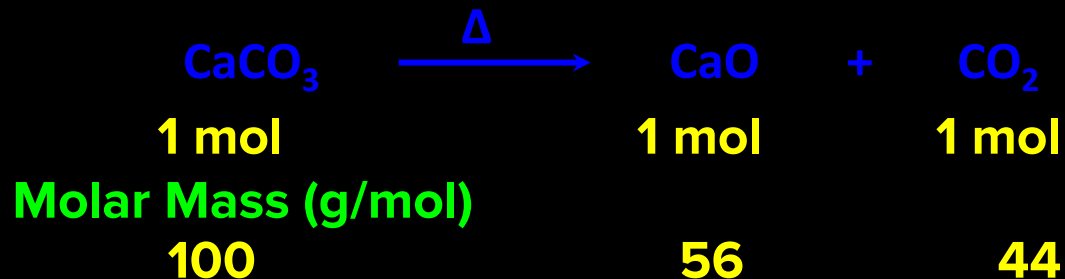
$(200 \times 10^3) \times \frac{95}{100} \text{ g} \longrightarrow ?$

$190 \times 10^3 \text{ g} \longrightarrow ?$

Let's Solve

Ex-1. Calculate the amount of lime (CaO) that can be prepared by heating 200 kg of limestone (CaCO₃) that is 95% pure.

Solution:



Converting Moles
into Mass

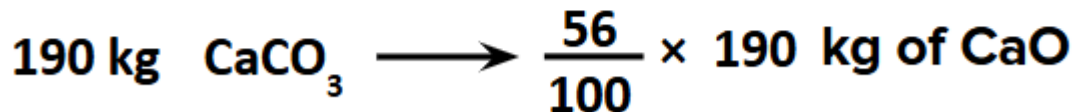
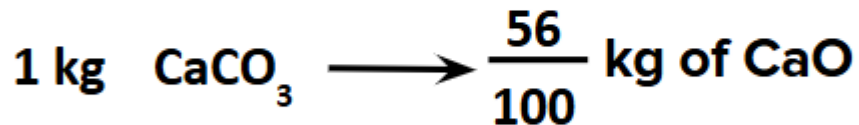
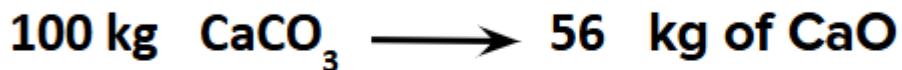
Let's Solve

Ex-1. Calculate the amount of lime (CaO) that can be prepared by heating 200 kg of limestone (CaCO₃) that is 95% pure.

Solution:

$$\text{Actual amount of CaCO}_3 = \frac{95}{100} \times 200 = 190 \text{ kg}$$

From Stoichiometric Calculations



$$= 106.4 \text{ kg of CaO}$$

What is Percentage Yield of Reaction ?

Possible reasons are ??

Sometimes, the reaction may not get completed according to initial amount taken.

In such cases, the efficiency of the process is measured using Percentage Yield

$$\% \text{ age Yield} = \frac{\text{Product Mass (Actual)}}{\text{Product Mass (Theoretical)}} \times 100$$

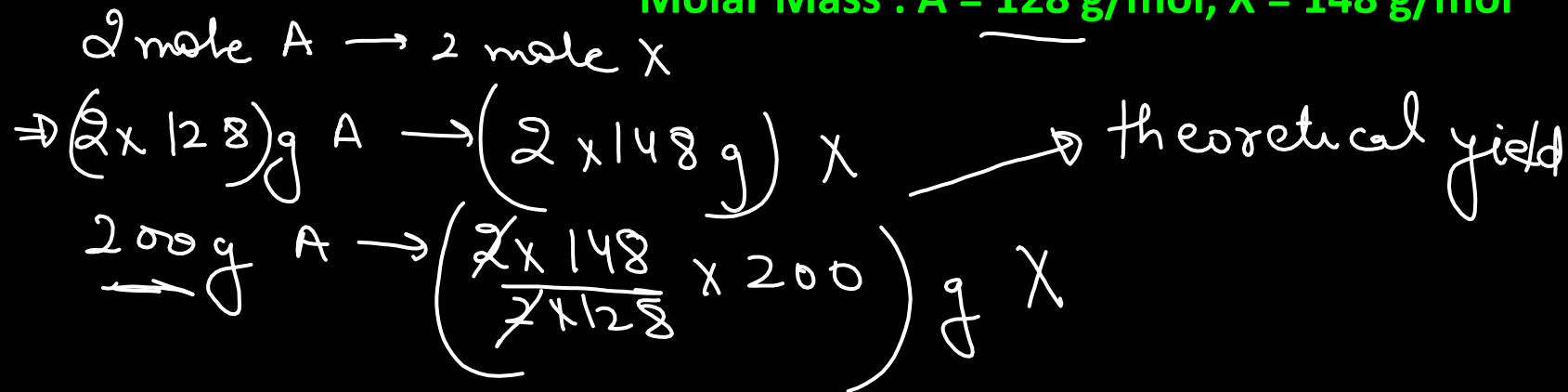
Let's Solve

Ex-3. A on controlled oxidation gives X according to the reaction:



Some of the reactant A oxidizes and only 70 % of maximum yield is obtained. What mass of X is produced if 200 g of A is taken ?

Molar Mass : A = 128 g/mol, X = 148 g/mol





2 mol

2 mol

1 mol

1 mol

128 g

148 g

128 g of A $\longrightarrow$ 148 g of X

1 g of A $\longrightarrow$ $\frac{148}{128}$ g of X

$\therefore$ 200 g A $\longrightarrow$ $\frac{148}{128} \times 200$ g of X

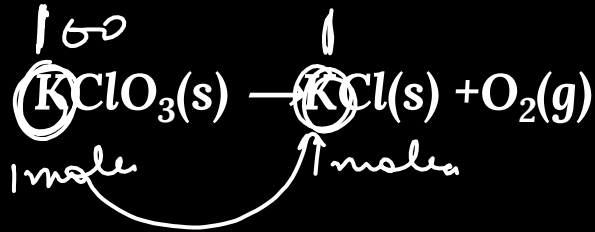
$$\% \text{ age Yield} = \frac{\text{Product Mass (Actual)}}{\text{Product Mass (Theoretical)}} \times 100$$

$$\text{Mass (Actual)} = \frac{\% \text{ age Yield}}{100} \times \text{Mass (Theoretical)}$$

$$\text{Mass (Actual)} = \frac{70}{100} \times \frac{148}{128} \times 200 \text{ g} = 161.875 \text{ g}$$

Principle of atomic conservation (POAC)

According to this principle as the number of atoms remain conserved during a chemical reaction the moles of atoms also remain conserved.

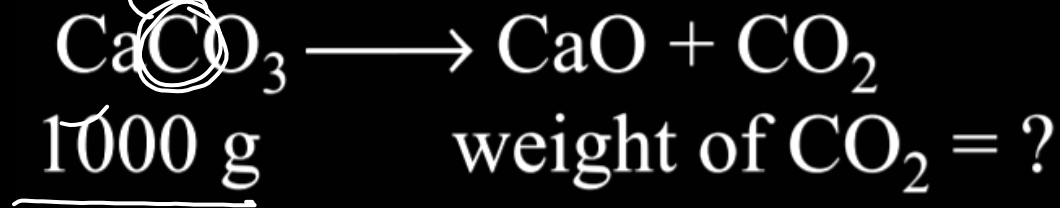


Applying POAC for K atoms

Moles of K atoms in KClO_3 = Moles of K atoms in KCl

$$1 \times n_{\text{KClO}_3} = 1 \times n_{\text{KCl}}$$

$$1 \times \frac{W_{\text{KClO}_3}}{M_{\text{KClO}_3}} = 1 \times \frac{W_{\text{KCl}}}{M_{\text{KCl}}}$$

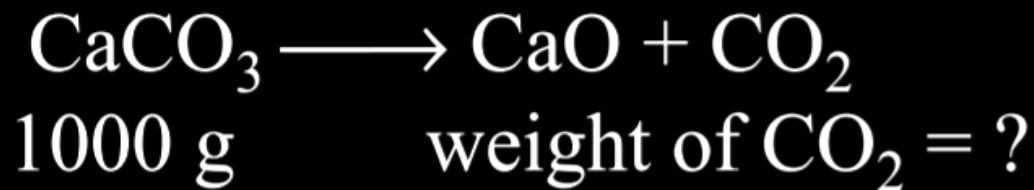


$$\text{moles of CaCO}_3 = \frac{1000}{100} = 10 \text{ mole CaCO}_3$$

$$= 10 \text{ mole C}$$

$$= 10 \text{ mole CO}_2$$

$$= 10 \times 44 \text{ g} = 440 \text{ g}$$



Solution:

POAC on C atom

$$1 \times \text{mole of CaCO}_3 = 1 \text{ mole of CO}_2$$

$$\frac{\text{Weight of CaCO}_3}{\text{MM of CaCO}_3} = \frac{\text{Weight of CO}_2}{\text{MM of CO}_2}$$

$$\frac{1000}{100} = \frac{\text{weight of CO}_2}{44}$$

$$\text{Weight of CO}_2 = 440 \text{ g}$$

Introduction of Solution



Solution

Solution is homogeneous mixture of two or more components.

Homogeneous

Mixture in which solute and solvent are in the same phase.



On the basis of number of components

Classification :

No. of components	Solution type
✓ 2	Binary solution
3	Ternary solution
4	Quaternary solution

Note :

For any n number of components, only one component will be solvent and rest $(n - 1)$ components are solutes.



On the basis of physical state of solvent

Classification :

Type of Solution	Solute	Solvent	Common Examples
<i>Gaseous Solutions</i>	Gas	Gas	Mixture of oxygen and nitrogen gases
	Liquid	Gas	Chloroform mixed with nitrogen gas
	Solid	Gas	Camphor in nitrogen gas
<i>Liquid Solutions</i>	Gas	Liquid	Oxygen dissolved in water
	Liquid	Liquid	Ethanol dissolved in water
	Solid	Liquid	Glucose dissolved in water
<i>Solid Solutions</i>	Gas	Solid	Solution of hydrogen in palladium
	Liquid	Solid	Amalgam of mercury with sodium
	Solid	Solid	Copper dissolved in gold

Mixture of **solute** + **solvent**

Minor
component

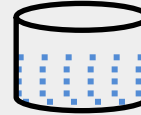
Eg. NaCl



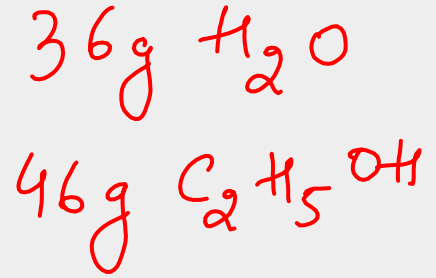
less moles
solute

Major
component

Eg. H₂O



more moles
solvent



$$\text{mole} = \frac{36}{18} = 2$$

H₂O

Clear
Solution

$$\text{mole} = \frac{46}{46} = 1$$

C₂H₅OH



Which of the following is/are not a solution?

Soda water

Smoke

Air

Benzene in water

Brass



ANSWER

Soda water

Smoke

Air

Benzene in water

Brass





Define Strength/Concentration of the solution

The amount of solute dissolved in per unit amount of solvent or solution is called strength of solution

Various methods of measuring strength of a solution

1) Mole fraction (X)

2) Mass by Percent (% w/w)

3) Volume by Percent (% v/v)

4) Mass by Volume Percent (% w/v)

5) Parts per million (ppm)

6) Molality, m

7) Molarity, M

1) Mole fraction (X)

A + B

$$X_A = \frac{n_A}{n_A + n_B}$$

$$X_B = \frac{n_B}{n_A + n_B}$$

Mole fraction, $X = \frac{\text{Moles of solute}}{\text{Total moles in solution}}$

$$X = \frac{n_{\text{solute}}}{n_{\text{solute}} + n_{\text{solvent}}}$$

If A, B, C... are different solutes and solvent

Then

$$X_A + X_B + X_C + \dots = 1$$

$$\sum_{i=A}^N X_i = 1$$

$$X_{C_2H_5OH} = \frac{1}{3}$$

$$\begin{cases} 36g H_2O \\ 46g C_2H_5OH \end{cases}$$

$$\text{Solute} = 0.3 = X$$

$$\text{Solvent} = 1 - 0.3 = 0.7$$

$$n_{H_2O} = \frac{36}{18} = 2$$

$$n_{C_2H_5OH} = \frac{46}{46} = 1$$

$$X_{H_2O} = \frac{n_{H_2O}}{n_{H_2O} + n_{C_2H_5OH}} = \frac{2}{1+2} = \frac{2}{3}$$

2) Mass by mass Percent (% w/w)

Imp - 9th syllabus

$$\% \text{ w/w} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$$

NaCl solution^{aq}
↓
solute

17% m/m

17 g NaCl (solute)
↓
100 g soln



3) Volume by volume Percent (% v/v)

$$\% \text{ v/v} = \frac{V_{\text{solute}}}{V_{\text{solution}}} \times 100$$

lig in lig soln
5% v/v alcohol
100ml → 5ml alcohol



4) Mass by Volume Percent (% w/v)

$$\% \text{ w/v} = \frac{W_{\text{solute}}}{V_{\text{solution}}} \times 100$$

18.0% w/v NaCl in water
18g solute $\rightarrow$ 100 ml soln

Define Parts per million (ppm) .

$$\text{million} = 10^6$$

5) Parts per million (ppm)

$$\text{Parts per million} = \frac{\text{Mass of solute}}{\text{Mass of solvent}} \times 10^6$$

very dilute soln
Solution
What does 4 ppm of solution means ?

4 ppm means 4g of solute is present in 10^6 g of ~~solvent~~.
Solution

4g solute $\rightarrow$ 10^6 g soln.

very dilute
soln

ppb

parts per billion

10^9

17 ppb

17g Cl_2 is
in 10^9 g of
pool water

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CONCENTRATION TERMS

Molality and Molarity

m

M



Molality (m)

It represents the moles of solute present in **1 kilogram** of **solvent**

$$m = \frac{\text{Moles of Solute (mol)}}{\text{Mass of Solvent (kg)}}$$

Unit : mol/kg or Molal

Example : 2.5 m aqueous solution of Glucose

2.5 moles of Glucose are present in 1 kg of H₂O

↓
Solute

$$m = \frac{n_{\text{solute}}}{W_{\text{solvent}} (\text{kg})}$$

6 m NaCl aq soln
6 moles NaCl → 1 Kg H₂O

$$m = \frac{n_{\text{solute}}}{W_{\text{solvent}}}$$

$$\Rightarrow m \times W_{\text{solvent}} = n_{\text{solute}}$$

Molarity (M)

It represents the moles of solute present in **1 litre** of **solution**.

$$M = \frac{\text{Moles of Solute (mol)}}{\text{Volume of Solution (L)}} \quad \text{Unit : mol/L or Molar}$$

Example : **1.7 M** NaCl aqueous solution

1 L of solution has 1.7 moles of NaCl

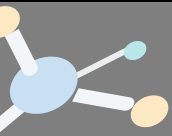
$$M = \frac{n_{\text{solute}}}{\text{Vol. of soln. (L)}}$$

$$n_{\text{solute}} = M \times V(\text{L})$$

Summary.



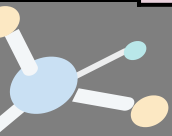
Conc. Term	Formula
Mass Percentage	$\%W/W = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$
Mass by Volume Percentage	$\%W/V = \frac{\text{Mass of solute}}{\text{Volume of solution}} \times 100$
Volume by volume percentage	$\%V/V = \frac{\text{Volume of solute (cm}^3\text{)}}{\text{Volume of solution (cm}^3\text{)}} \times 100$
Molarity	$\text{Molarity, (M)} = \frac{\text{Moles of solute}}{\text{Volume of solution (L)}}$
Molality	$\text{Molality, (m)} = \frac{\text{Moles of solute}}{\text{Mass of solvent (Kg)}}$



Summary.



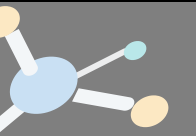
Conc. Term	Formula
Mole Fraction	$\text{Mole fraction} = \frac{\text{Moles of component}}{\text{Total number of moles present in solution}}$ $x_A = \frac{n_A}{n_A + n_B + n_C + \dots}$
Parts per million	$\text{ppm of a solute in solution} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 10^6$
Degree of hardness of water	$\text{Hardness of water} = \frac{\text{mass of CaCO}_3}{\text{Total mass of water}} \times 10^6$



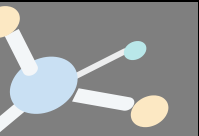
Concentration Terms



Temperature independent	Temperature dependent
Mass percentage (%w/w)	Volume percentage (%v/v)
Mole fraction	Mass by volume Percentage (%w/v)
Molality (m)	Molarity (M)
Parts per million (ppm)	



Dilution Formula



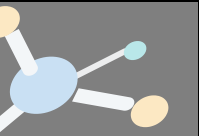
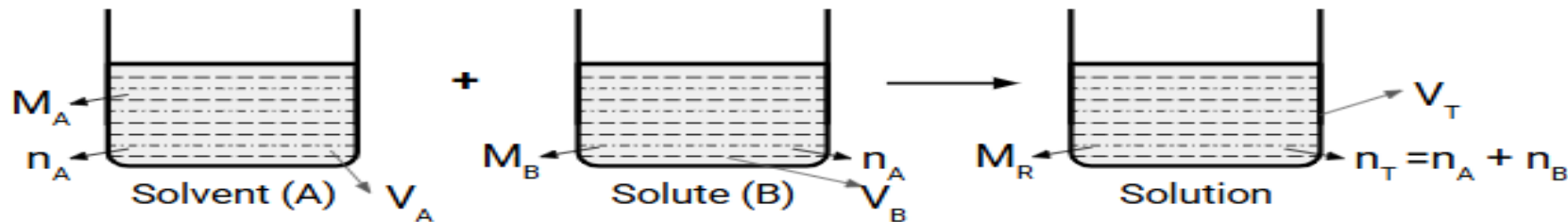
Concentration Terms



Molarity of Mixing

Condition:

- 1) Compound should be of same nature
- 2) If ionic compounds are given then at least one ion should be same.



Concentration Terms

Some important relations

$$M = \frac{10 \times \%w/v}{M_{\text{solute}}}$$

$$M = \frac{10 \times \%w/w \times d}{M_{\text{solute}}}$$

$$\% w/v = \% w/w \times d$$

$$m = \frac{X_{\text{solute}}}{X_{\text{solvent}}} \times \frac{1000}{M_{\text{solvent}}}$$

$$m = \frac{1000 \times M}{1000d - M}$$

Where,

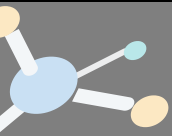
M = Molarity

m = Molality

d = Density

M_{Solvent} = Molar mass of solvent

M_{solute} = Molar mass of solute



NSEJS IMPORTANT QUESTIONS

Relationship between molality and molarity

9 mp

$$\frac{1}{m} = \frac{\rho}{M} - \frac{m_B}{1000}$$

Where, ρ = density of solution
 m = molality
 M = Molarity
 m_B = molar mass of solute

$$M = \frac{\rho \times m/m\% \times 10}{m_B}$$

Q. An aqueous solution of ethanol has density 1.025 g/mL and it is 2 M. What is the molality of this solution ?

A 1.79

B 2.143

C 1.951

D None of these



$$m = ?$$

Q. An aqueous solution of ethanol has density 1.025 g/mL and it is 2 M. What is the molality of this solution ?

A

1.79

B

2.143

C

1.951

D

None of these

Q. Calculate molarity of KI If the density of 20% (mass/mass) aqueous KI is 1.202 g ml⁻¹.

A

3.50

C

2.50

B

1.45

D

5.45

$$M = \frac{\rho \times \frac{m}{m^*} \times 10}{m_B}$$

Q. Calculate molarity of KI If the density of 20% (mass/mass) aqueous KI is 1.202 g ml^{-1} .

A

3.50

C

2.50

B

1.45

D

5.45

Let's Solve

Q. How many grams of a 5 % w/w NaCl solution are necessary to yield 3.2 g NaCl ?


$$\begin{aligned} 5 \text{ g NaCl} &\rightarrow 100 \text{ g soln} \\ 3.2 \text{ g} &\rightarrow \left(\frac{100}{5} \times 3.2 \right) \text{ g} \end{aligned}$$

Let's Solve

Q. How many grams of a 5 % w/w NaCl solution are necessary to yield 3.2 g NaCl ?

$$\% \text{ w/w} = \frac{\text{w}_{\text{solute}}}{\text{w}_{\text{solution}}} \times 100$$

$$5 = \frac{3.2 \text{ g}}{\text{w}_{\text{solution}}} \times 100$$

$$\text{w}_{\text{solution}} = \frac{3.2 \text{ g} \times 100}{5} = 64 \text{ g}$$

Let's Solve

Q. A solution of urea is 3000 ppm by weight. If density of the solution is 1.04 g/mL, determine the molar strength of

H W

MM of Urea (NH_2CONH_2) = 60 g/mol

Let's Solve

Q. A solution of urea is 3000 ppm by weight. If density of the solution is 1.04 g/mL, determine the molar strength of solution.

Solution

MM of Urea (NH_2CONH_2) = 60 g/mol

10^6 g of solution contains 3000 g of Urea

ppm Interpretation

Density Interpretation

$$V_{\text{solution}} = \frac{10^6 \text{ g}}{1.04 \text{ g/mL}} = \frac{10^6}{1.04} \text{ mL} = \frac{10^3}{1.04} \text{ L}$$

$$n_{\text{urea}} = \frac{3000 \text{ g}}{60 \text{ g/mol}} = 50 \text{ mol}$$

$$M = \frac{50 \times 1.04 \text{ mol}}{10^3 \text{ L}} = \mathbf{0.052 \text{ M}}$$

Q: chalk is composed of calcium carbonate, CaCO_3 . This water-insoluble compound is formed when a solution of calcium chloride is added to a solution of sodium carbonate. How many milliliters of 0.25 M calcium chloride are needed to react completely with 50 mL of 0.15 M Na_2CO_3 solution?

IJSO 2016, Indonesia

A 10

B 15

C 30

D 60

Q-2: Chalk is composed of calcium carbonate, CaCO_3 . This water-insoluble compound is formed when a solution of calcium chloride is added to a solution of sodium carbonate. How many milliliters of 0.25 M calcium chloride are needed to react completely with 50 mL of 0.15 M Na_2CO_3 solution?

IJSO 2016, Indonesia

Q-2: Chalk is composed of calcium carbonate, CaCO_3 . This water-insoluble compound is formed when a solution of calcium chloride is added to a solution of sodium carbonate. How many milliliters of 0.25 M calcium chloride are needed to react completely with 50 mL of 0.15 M Na_2CO_3 solution?

IJSO 2016, Indonesia

A

10

B

15

C

30

D

60

Solution:

The reaction is $\text{CaCl}_2(\text{aq}) + \text{Na}_2\text{CO}_3(\text{aq}) \rightarrow \text{CaCO}_3(\text{s}) + 2\text{NaCl}(\text{aq})$

Mol of $\text{Na}_2\text{CO}_3 = 50 \text{ mL} \times 0.15 \text{ M} = 7.5 \times 10^{-3} \text{ mol}$

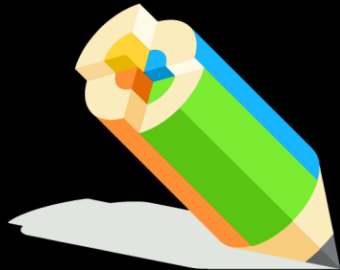
Mol of $\text{CaCl}_2 = \text{Mol of } \text{Na}_2\text{CO}_3 = 7.5 \times 10^{-3} \text{ mol}$

Volume of $\text{CaCl}_2 = 7.5 \times 10^{-3} \text{ mol} / 0.25 \text{ M} = 0.03 \text{ L} = 30 \text{ mL}$

Dilution and Mixing of Solutions



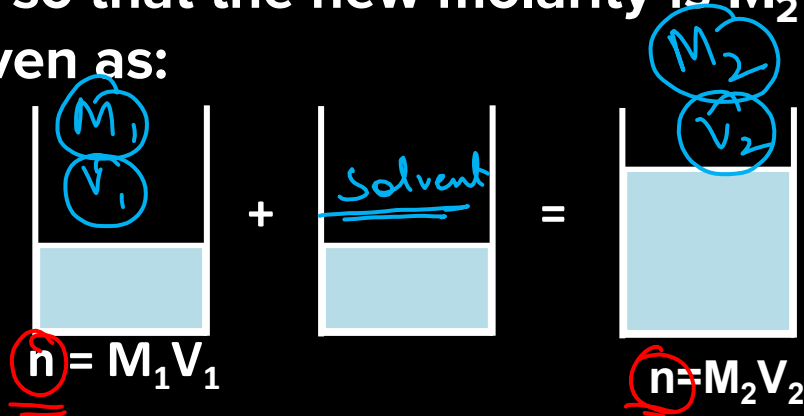
**What happens when we mix solutions
and define Dilution law and Mixing law?**



$$M = \frac{n}{V}$$

Dilution Law :

If a solution having molarity M_1 and Volume V_1 is diluted to Volume V_2 so that the new molarity is M_2 , then the molarity equation is given as:



$$M_1 V_1 = M_2 V_2$$

Dilution Law

$$n = M \times V$$

V_1 = Initial volume

V_2 = Final volume

Total number of moles in the solution remains same.

Mixing Law

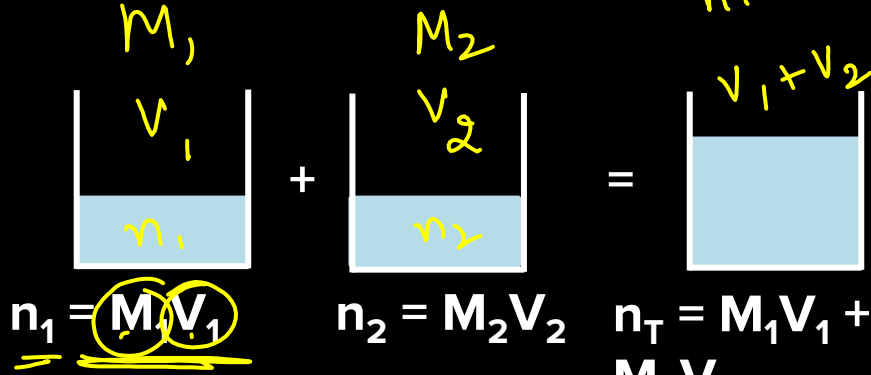
$$M = \frac{n}{V}$$

$$\frac{n_1 + n_2}{V_1 + V_2}$$

$$M_f = \frac{n_1 + n_2}{V_1 + V_2}$$

$$M = \frac{n}{V}$$

$$n = M \times V$$



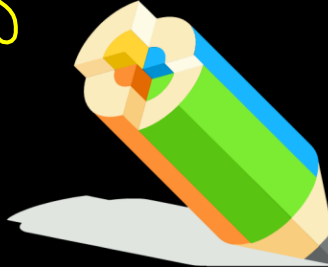
$$M_f = \frac{\text{Moles of solute}}{\text{Volume of solution}} = \frac{M_1 V_1 + M_2 V_2}{V_1 + V_2}$$

$$= \frac{\text{total moles of solute}}{\text{total vol of soln}}$$

$\frac{n_1 + n_2}{V_1 + V_2}$

$$M_f = \frac{M_1 V_1 + M_2 V_2 + \dots + M_n V_n}{V_1 + V_2 + \dots + V_n}$$

n = moles



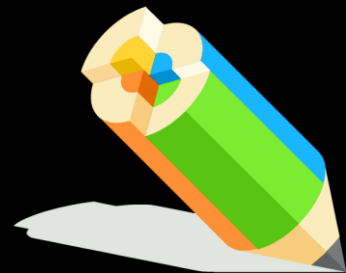
Let's Solve

Calculate the molarity of the following solution..

(a) 30 ml of 0.5M H_2SO_4 diluted to 500 ml

$$M_1 V_1 = M_2 V_2$$

$$\Rightarrow 0.5 \times \underline{30} = \underset{?}{\textcircled{M_2}} \times \underline{500}$$



Let's Solve

Calculate the molarity of the following solution..

(a) 30 ml of 0.5M H_2SO_4 diluted to 500 ml

Solution:

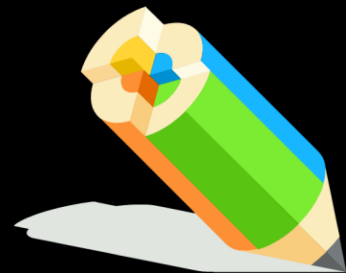
$$M_1 V_1 = M_2 V_2$$

$$0.5 \times 30 = M_2 \times 500$$

$$M_2 = \frac{30 \times 0.5}{500}$$

$\therefore$ New molarity, $M_2 = 0.03 \text{ M}$

*On Dilution, number of
moles
 $n = MV$ remains same*



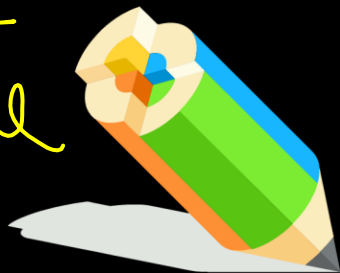
Q. Find final molarity in each case.

(i) 500 mL 0.1M HCl + 500 mL 0.2 M HCl

(ii) 50mL 0.1 M HCl + 150 mL 0.3M HCl + 300mL H₂O

$$\textcircled{a} \quad \frac{M_1 V_1 + M_2 V_2}{V_1 + V_2} = M_f$$

$$\textcircled{b} \quad M_f = \frac{\text{total moles of solute}}{\text{total of soln (lt)}} = \frac{M_1 V_1 + M_2 V_2}{V_{\text{total}}}$$



Q. Find final molarity in each case.

(i) 500 mL 0.1M HCl + 500 mL 0.2 M HCl

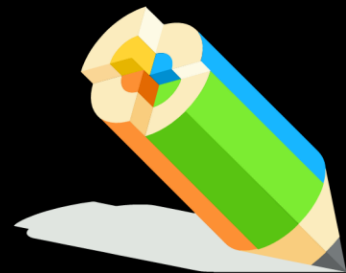
(ii) 50mL 0.1 M HCl + 150 mL 0.3M HCl + 300mL H₂O

Sol: (i) 500mL 0.1M HCl + 500mL 0.2M HCl

$$M_f = \frac{(500 \times 0.1) + (500 \times 0.2)}{500 + 500} = 0.15M$$

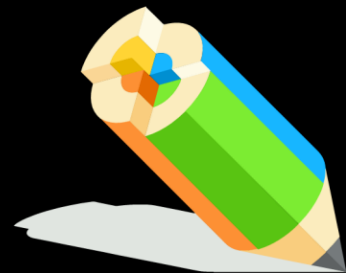
(ii) 50mL 0.1M HCl + 150 mL 0.3M HCl + 300 mL H₂O

$$M_f = \frac{(50 \times 0.1) + (150 \times 0.3)}{50 + 150 + 300} = \frac{50}{500} = 0.1M$$



Let's Solve

Calculate the final concentration of solute after 250 mL of 0.5 M antifreeze solution and 500 mL of 4.0 M antifreeze solution are combined and diluted to 2.0 L with water.



Let's Solve

Calculate the final concentration of solute after 250 mL of 0.5 M antifreeze solution and 500 mL of 4.0 M antifreeze solution are combined and diluted to 2.0 L with water.

Solution:

$$M_1 = 0.5 \text{ M}$$

$$M_2 = 4.0 \text{ M}$$

$$V_1 = 250 \text{ mL}$$

$$V_2 = 500 \text{ mL}$$

$$\text{Final Volume}(V_f) = 2 \text{ L} = 2000 \text{ mL}$$

$$M_f = \frac{M_1 V_1 + M_2 V_2}{V_f}$$

$$= \frac{0.5 \times 250 + 4 \times 500}{2000}$$

$$= \frac{125 + 2000}{2000} = 1.0625 \text{ M}$$



Q. How much volume of 10M HCl should be diluted with water to prepare 2.00 L of 5M HCl?

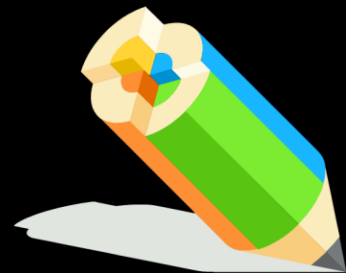
2 L

5 M

$$M_1 V_1 = M_2 V_2$$

$$10 \times \textcircled{V_1} = 5 \times 2$$

$$\text{moles of HCl} = 2 \times 5 = \underline{10 \text{ moles}}$$



Q. How much volume of 10M HCl should be diluted with water to prepare 2.00 L of 5M HCl?

Sol: In dilution the following equation is applicable :

$$\begin{aligned}M_1V_1 &= M_2V_2 \\10 \times V_1 &= 5 \times 2.00 \\V_1 &= \frac{5 \times 2.00}{10} = 1.00\text{L}\end{aligned}$$



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- **REGULAR NSEJS PATTERN TESTS**
- **HARD COPY NSEJS TATVA**
- **Recordings for Basics Coverage (P/C/B/M)**
- **ALL INDIA NSEJS MOCK TEST SERIES 2026**
- **Batch Start Date: 7th April, 2026**

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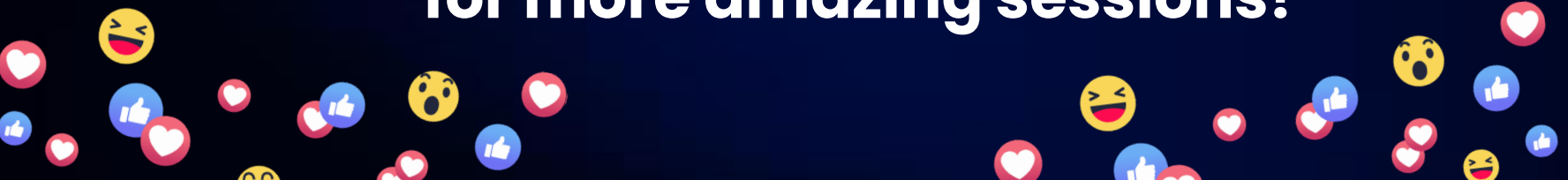
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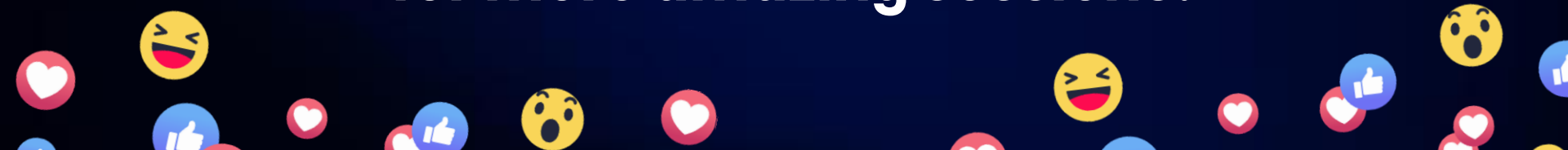


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